# Instituto Tecnológico de Costa Rica

## Operations Research - Semester II

## Simplex

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## Simplex

The simplex algorithm was invented by Robert Dantzig in 1947 to solve Linear Programming problems. Dantzig proved that the feasible region (area that satisfies all constraints) of a Linear Programming problem forms a convex set. He also proved that the optimal solution of a LP problem is located at a vertex of the convex body of the feasible region.

Dantzig then used this theorems to create Simplex, an algorithm that moves through the vertices of the feasible convex body until finding the optimal solution to the problem. Simplex is a mechanical method that uses matrix operations to determine the solution.

To setup the simplex algorithm we will:

- For each $\leq$ constraint add a slack variable $s_i$
- For each $\geq$ constraint add an excess variable $e_i$ and an artificial variable $a_i$.
- for each $=$ constraint add an artificial variable $a_i$
- for each artificial variable added:
  - Add $Ma_i$ to the objective function if minimizing.
  - Add $-Ma_i$ to the objective function if maximizing.
  - M is a constant of a large value, but not infinite.

### Example

$$\begin{gathered}\text{Maximize Z} = x_1 + x_2 \\ x_1 + 2x_2 \leq 3 \\ 4x_1 + 5x_2 \geq 6 \\ 7x_1 + 8x_2 = 9\end{gathered}$$

Converts to

$$\begin{gathered}\text{Maximize Z} = x_1 + x_2 - Ma_1 - Ma_2 \\ x_1 + 2x_2 + s_1 \leq 3 \\ 4x_1 + 5x_2 + e_1 + a_1 \geq 6 \\ 7x_1 + 8x_2 + a_2 = 9\end{gathered}$$

### Simplex Table

Using the same example as above, we should construct a matrix called Simplex Table as follows:

| Z | $x_1$ | $x_2$ | $s_1$ | $e_1$ | $a_1$ | $a_2$ | b |
|---|---|---|---|---|---|---|---|
| 1 | -1 | -1 | 0 | 0 | M | M | 0 |
| 0 | 1 | 2 | 1 | 0 | 0 | 0 | 3 |
| 0 | 4 | 5 | 0 | -1 | 1 | 0 | 6 |
| 0 | 7 | 8 | 0 | 0 | 0 | 1 | 9 |

Execution of Simplex algorithm:

1. If there are any artificial variables, our first step should be to remove the Ms in their columns through matrix operations, leaving canonical vectors on artificial variable columns.

2. if there are any negative
textbackslash positive values in the first row (excluding Z and b) when maximizing textbackslash minimizing (respectively)

   (a) Identify the pivot

      i. Indetify a column X with the most negative
      textbackslash positive (maximizing
      textbackslash minimizing), divide all values on column b with their respective value on X.

      ii. Exclude 0s and negative values

      iii. The pivot will be the cell on column X on the row with the **smallest** division result.

   (b) Through matrix operations, create a canonical vector on the column of the pivot, leaving 1 exactly on the pivot cell.

   (c) Go to 2

3. The simplex table will contain the optimal solution.

   - In column b, the first cell shows the value of Z.
   - Every variable with a non-canonical vector in their column has a value of 0.
   - Variables with a canonical vector in their column have a value of what the b column has on the row where they have a 1.

# Dakota (Multiples)

Maximize

$$Z = 60.00x_1 + 35.00x_2 + 20.00x_3$$

Subject to

$$8.00x_1 + 6.00x_2 + 1.00x_3 \leq 48.00$$
$$4.00x_1 + 2.00x_2 + 1.50x_3 \leq 20.00$$
$$2.00x_1 + 1.50x_2 + 0.50x_3 \leq 8.00$$
$$0.00x_1 + 2.00x_2 + 0.00x_3 \leq 5.00$$

Simplex Table

| | Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|---|
| | 1 | -60 | -35 | -20 | 0 | 0 | 0 | 0 | 0 |
| | 0 | 8 | 6 | 1 | 1 | 0 | 0 | 0 | 48 |
| | 0 | 4 | 2 | 1.5 | 0 | 1 | 0 | 0 | 20 |
| | 0 | 2 | 1.5 | 0.5 | 0 | 0 | 1 | 0 | 8 |
| | 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

# Intermediate Tables

## Pivoting 1

### Most Negative

Column 2 (-60)

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | -60 | -35 | -20 | 0 | 0 | 0 | 0 | 0 |
| 0 | 8 | 6 | 1 | 1 | 0 | 0 | 0 | 48 |
| 0 | 4 | 2 | 1.5 | 0 | 1 | 0 | 0 | 20 |
| 0 | 2 | 1.5 | 0.5 | 0 | 0 | 1 | 0 | 8 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

### Fractions

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b | Frac |
|---|---|---|---|---|---|---|---|---|---|
| 1 | -60 | -35 | -20 | 0 | 0 | 0 | 0 | 0 | |
| 0 | 8 | 6 | 1 | 1 | 0 | 0 | 0 | 48 | 6.00 |
| 0 | 4 | 2 | 1.5 | 0 | 1 | 0 | 0 | 20 | 5.00 |
| 0 | 2 | 1.5 | 0.5 | 0 | 0 | 1 | 0 | 8 | 4.00 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 | inf |

$48.00/8.00 = 6.00$
$20.00/4.00 = 5.00$
$8.00/2.00 = 4.00$
$5.00/0.00 = inf$
Smallest fraction: 4.00 $\rightarrow$ Pivot: row 4

### Pivot

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | -60 | -35 | -20 | 0 | 0 | 0 | 0 | 0 |
| 0 | 8 | 6 | 1 | 1 | 0 | 0 | 0 | 48 |
| 0 | 4 | 2 | 1.5 | 0 | 1 | 0 | 0 | 20 |
| 0 | 2 | 1.5 | 0.5 | 0 | 0 | 1 | 0 | 8 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

### Canonization

$R_4 \leftarrow R_4/2.00$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | -60 | -35 | -20 | 0 | 0 | 0 | 0 | 0 |
| 0 | 8 | 6 | 1 | 1 | 0 | 0 | 0 | 48 |
| 0 | 4 | 2 | 1.5 | 0 | 1 | 0 | 0 | 20 |
| 0 | 1 | 0.75 | 0.25 | 0 | 0 | 0.5 | 0 | 4 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

$R_1 \leftarrow R_1 + 60.00R_4$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 10 | -5 | 0 | 0 | 30 | 0 | 240 |
| 0 | 8 | 6 | 1 | 1 | 0 | 0 | 0 | 48 |
| 0 | 4 | 2 | 1.5 | 0 | 1 | 0 | 0 | 20 |
| 0 | 1 | 0.75 | 0.25 | 0 | 0 | 0.5 | 0 | 4 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

$R_2 \leftarrow R_2 + -8.00R_4$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 10 | -5 | 0 | 0 | 30 | 0 | 240 |
| 0 | 0 | 0 | -1 | 1 | 0 | -4 | 0 | 16 |
| 0 | 4 | 2 | 1.5 | 0 | 1 | 0 | 0 | 20 |
| 0 | 1 | 0.75 | 0.25 | 0 | 0 | 0.5 | 0 | 4 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

$R_3 \leftarrow R_3 + -4.00R_4$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 10 | -5 | 0 | 0 | 30 | 0 | 240 |
| 0 | 0 | 0 | -1 | 1 | 0 | -4 | 0 | 16 |
| 0 | 0 | -1 | 0.5 | 0 | 1 | -2 | 0 | 4 |
| 0 | 1 | 0.75 | 0.25 | 0 | 0 | 0.5 | 0 | 4 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

$R_5 \leftarrow R_5 + -0.00R_4$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 10 | -5 | 0 | 0 | 30 | 0 | 240 |
| 0 | 0 | 0 | -1 | 1 | 0 | -4 | 0 | 16 |
| 0 | 0 | -1 | 0.5 | 0 | 1 | -2 | 0 | 4 |
| 0 | 1 | 0.75 | 0.25 | 0 | 0 | 0.5 | 0 | 4 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

**Pivot Result**

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 10 | -5 | 0 | 0 | 30 | 0 | 240 |
| 0 | 0 | 0 | -1 | 1 | 0 | -4 | 0 | 16 |
| 0 | 0 | -1 | 0.5 | 0 | 1 | -2 | 0 | 4 |
| 0 | 1 | 0.75 | 0.25 | 0 | 0 | 0.5 | 0 | 4 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

## Pivoting 2

### Most Negative

Column 4 (-5)

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 10 | -5 | 0 | 0 | 30 | 0 | 240 |
| 0 | 0 | 0 | -1 | 1 | 0 | -4 | 0 | 16 |
| 0 | 0 | -1 | 0.5 | 0 | 1 | -2 | 0 | 4 |
| 0 | 1 | 0.75 | 0.25 | 0 | 0 | 0.5 | 0 | 4 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

### Fractions

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b | Frac |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 10 | -5 | 0 | 0 | 30 | 0 | 240 | |
| 0 | 0 | 0 | -1 | 1 | 0 | -4 | 0 | 16 | -16.00 |
| 0 | 0 | -1 | 0.5 | 0 | 1 | -2 | 0 | 4 | 8.00 |
| 0 | 1 | 0.75 | 0.25 | 0 | 0 | 0.5 | 0 | 4 | 16.00 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 | inf |

$16.00/-1.00 = -16.00$
$4.00/0.50 = 8.00$
$4.00/0.25 = 16.00$
$5.00/0.00 = inf$
Smallest fraction: 8.00 $\rightarrow$ Pivot: row 3

### Pivot

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 10 | -5 | 0 | 0 | 30 | 0 | 240 |
| 0 | 0 | 0 | -1 | 1 | 0 | -4 | 0 | 16 |
| 0 | 0 | -1 | 0.5 | 0 | 1 | -2 | 0 | 4 |
| 0 | 1 | 0.75 | 0.25 | 0 | 0 | 0.5 | 0 | 4 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

### Canonization

$R_3 \leftarrow R_3/0.50$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 10 | -5 | 0 | 0 | 30 | 0 | 240 |
| 0 | 0 | 0 | -1 | 1 | 0 | -4 | 0 | 16 |
| 0 | 0 | -2 | 1 | 0 | 2 | -4 | 0 | 8 |
| 0 | 1 | 0.75 | 0.25 | 0 | 0 | 0.5 | 0 | 4 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

$R_1 \leftarrow R_1 + 5.00R_3$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 0 | 0 | -1 | 1 | 0 | -4 | 0 | 16 |
| 0 | 0 | -2 | 1 | 0 | 2 | -4 | 0 | 8 |
| 0 | 1 | 0.75 | 0.25 | 0 | 0 | 0.5 | 0 | 4 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

$R_2 \leftarrow R_2 + 1.00R_3$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 0 | -2 | 0 | 1 | 2 | -8 | 0 | 24 |
| 0 | 0 | -2 | 1 | 0 | 2 | -4 | 0 | 8 |
| 0 | 1 | 0.75 | 0.25 | 0 | 0 | 0.5 | 0 | 4 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

$R_4 \leftarrow R_4 + -0.25R_3$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 0 | -2 | 0 | 1 | 2 | -8 | 0 | 24 |
| 0 | 0 | -2 | 1 | 0 | 2 | -4 | 0 | 8 |
| 0 | 1 | 1.25 | 0 | 0 | -0.5 | 1.5 | 0 | 2 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

$R_5 \leftarrow R_5 + -0.00R_3$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 0 | -2 | 0 | 1 | 2 | -8 | 0 | 24 |
| 0 | 0 | -2 | 1 | 0 | 2 | -4 | 0 | 8 |
| 0 | 1 | 1.25 | 0 | 0 | -0.5 | 1.5 | 0 | 2 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

### Pivot Result

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 0 | -2 | 0 | 1 | 2 | -8 | 0 | 24 |
| 0 | 0 | -2 | 1 | 0 | 2 | -4 | 0 | 8 |
| 0 | 1 | 1.25 | 0 | 0 | -0.5 | 1.5 | 0 | 2 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

## Results

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 0 | -2 | 0 | 1 | 2 | -8 | 0 | 24 |
| 0 | 0 | -2 | 1 | 0 | 2 | -4 | 0 | 8 |
| 0 | 1 | 1.25 | 0 | 0 | -0.5 | 1.5 | 0 | 2 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

### Objective Value

Z = 280.00

### Variables

$x_1 = 2.00$
$x_2 = 0.00$
$x_3 = 8.00$

### Slack or Surplus

$s_1 = 24.00$
$s_2 = 0.00$
$s_3 = 0.00$
$s_4 = 5.00$

### Excess

# Multiple Solutions

A non-basic variable has a 0 on its first row, allowing us to pivot again and find another optimal solution.

## Pivoting 3

### Most Negative

Column 3 (0)

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 0 | -2 | 0 | 1 | 2 | -8 | 0 | 24 |
| 0 | 0 | -2 | 1 | 0 | 2 | -4 | 0 | 8 |
| 0 | 1 | 1.25 | 0 | 0 | -0.5 | 1.5 | 0 | 2 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

### Fractions

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b | Frac |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 | |
| 0 | 0 | -2 | 0 | 1 | 2 | -8 | 0 | 24 | -12.00 |
| 0 | 0 | -2 | 1 | 0 | 2 | -4 | 0 | 8 | -4.00 |
| 0 | 1 | 1.25 | 0 | 0 | -0.5 | 1.5 | 0 | 2 | 1.60 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 | 2.50 |

$24.00/-2.00 = -12.00$
$8.00/-2.00 = -4.00$
$2.00/1.25 = 1.60$
$5.00/2.00 = 2.50$
Smallest fraction: 1.60 $\rightarrow$ Pivot: row 4

### Pivot

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 0 | -2 | 0 | 1 | 2 | -8 | 0 | 24 |
| 0 | 0 | -2 | 1 | 0 | 2 | -4 | 0 | 8 |
| 0 | 1 | 1.25 | 0 | 0 | -0.5 | 1.5 | 0 | 2 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

### Canonization

$R_4 \leftarrow R_4/1.25$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 0 | -2 | 0 | 1 | 2 | -8 | 0 | 24 |
| 0 | 0 | -2 | 1 | 0 | 2 | -4 | 0 | 8 |
| 0 | 0.8 | 1 | 0 | 0 | -0.4 | 1.2 | 0 | 1.6 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

$R_1 \leftarrow R_1 + -0.00R_4$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 0 | -2 | 0 | 1 | 2 | -8 | 0 | 24 |
| 0 | 0 | -2 | 1 | 0 | 2 | -4 | 0 | 8 |
| 0 | 0.8 | 1 | 0 | 0 | -0.4 | 1.2 | 0 | 1.6 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

$R_2 \leftarrow R_2 + 2.00R_4$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 1.6 | 0 | 0 | 1 | 1.2 | -5.6 | 0 | 27.2 |
| 0 | 0 | -2 | 1 | 0 | 2 | -4 | 0 | 8 |
| 0 | 0.8 | 1 | 0 | 0 | -0.4 | 1.2 | 0 | 1.6 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

$R_3 \leftarrow R_3 + 2.00R_4$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 1.6 | 0 | 0 | 1 | 1.2 | -5.6 | 0 | 27.2 |
| 0 | 1.6 | 0 | 1 | 0 | 1.2 | -1.6 | 0 | 11.2 |
| 0 | 0.8 | 1 | 0 | 0 | -0.4 | 1.2 | 0 | 1.6 |
| 0 | 0 | 2 | 0 | 0 | 0 | 0 | 1 | 5 |

$R_5 \leftarrow R_5 + -2.00R_4$

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 1.6 | 0 | 0 | 1 | 1.2 | -5.6 | 0 | 27.2 |
| 0 | 1.6 | 0 | 1 | 0 | 1.2 | -1.6 | 0 | 11.2 |
| 0 | 0.8 | 1 | 0 | 0 | -0.4 | 1.2 | 0 | 1.6 |
| 0 | -1.6 | 0 | 0 | 0 | 0.8 | -2.4 | 1 | 1.8 |

**Pivot Result**

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 1.6 | 0 | 0 | 1 | 1.2 | -5.6 | 0 | 27.2 |
| 0 | 1.6 | 0 | 1 | 0 | 1.2 | -1.6 | 0 | 11.2 |
| 0 | 0.8 | 1 | 0 | 0 | -0.4 | 1.2 | 0 | 1.6 |
| 0 | -1.6 | 0 | 0 | 0 | 0.8 | -2.4 | 1 | 1.8 |

# Results

| Z | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $s_4$ | b |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 10 | 10 | 0 | 280 |
| 0 | 1.6 | 0 | 0 | 1 | 1.2 | -5.6 | 0 | 27.2 |
| 0 | 1.6 | 0 | 1 | 0 | 1.2 | -1.6 | 0 | 11.2 |
| 0 | 0.8 | 1 | 0 | 0 | -0.4 | 1.2 | 0 | 1.6 |
| 0 | -1.6 | 0 | 0 | 0 | 0.8 | -2.4 | 1 | 1.8 |

## Objective Value

Z = 280.00

## Variables

$x_1 = 0.00$
$x_2 = 1.60$
$x_3 = 11.20$

## Slack or Surplus

$s_1 = 27.20$
$s_2 = 0.00$
$s_3 = 0.00$
$s_4 = 1.80$

## Excess

## More Solutions

$a \times s1 + (1-a) \times s2$
Multiple solution 1($a = 0.30$):
$x_1 = 0.60$
$x_2 = 1.12$
$x_3 = 10.24$
Multiple solution 2($a = 0.50$):
$x_1 = 1.00$
$x_2 = 0.80$
$x_3 = 9.60$
Multiple solution 3($a = 0.70$):
$x_1 = 1.40$
$x_2 = 0.48$
$x_3 = 8.96$